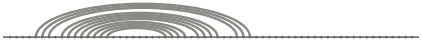
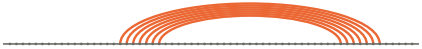
A What is conditioned on

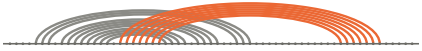
Scaffold G: input pairs without crossings



Extension G': added pairs without crossings, sharing no base with G

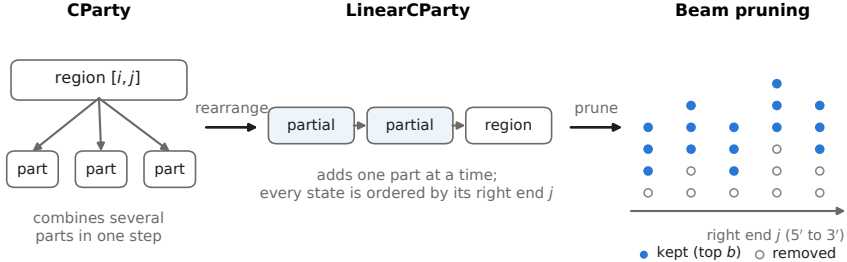


Completed structure $R = G \cup G'$



The ensemble holds every R that keeps G: $P(R \mid S, G) = w(R) / Z(S, G)$, where $w(R)$ is the Boltzmann weight and $Z(S, G)$ sums $w(R)$ over all such R.

B What is approximated



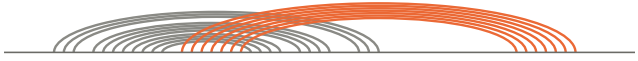
Ranking: a candidate gets its inside weight plus a rough estimate of its outside weight, taken from a preliminary run with a small beam. The top b candidates are kept for each state type and right end j. Local limits h (length of a new hairpin) and d (depth of scaffold branching) bound the candidates; here $h = d = \max(b, 30)$. Without the beam and these limits, the same calculation is exact.

C What is computed (bpRNA_PDB_345, 63 nt; exact; G = nested pairs of the reference structure)

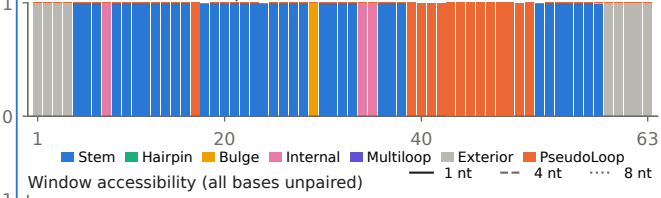
pair in G · new pair, no crossing · new pair crossing G

From one retained ensemble (the same probabilities are shared by all of these)

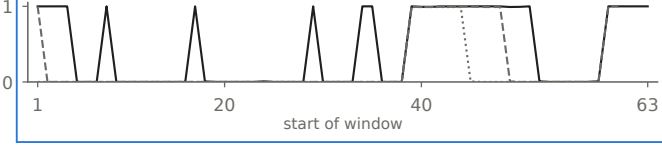
Base-pairing probabilities (darker = higher)



Seven structural-context probabilities



Window accessibility (all bases unpaired)



Boltzmann sampling: 1,000 structures

```
                                                                  count
GGAGGUCAGGGUCAGGAGCCCCCCCCUGAACCCAGGAUAACCCUCAAAGUCGGGGGGCAACCC
....(((.(((((((.[[[[[[[[)))).))))..)))............]]]]]]].....   975
....(((.(((((((.[[[[[.))))).))))..)))............]]]]]].....     8
....(((.(((((((..[[[[[[[[)))).))))..)))............]]]]]].....     6
```

Maximum expected accuracy structure ($\gamma = 1$)

```
....(((.(((((((.[[[[[[[[)))).))))..)))............]]]]]]].....
```

Same decomposition, separate maximization

Minimum-free-energy and suboptimal structures

```
                                                                  kcal/mol
....(((.(((((((.[[[[[[[[)))).))))..)))............]]]]]]].....    -29.4
....(((.(((((((.[[[[[.))))).))))..)))............]]]]]].....     -26.3
....(((.(((((((..[[[[[[[[)))).))))..)))............]]]]]].....    -26.3
....(((.(((((((.[[[[[[[[)))).))))..))).((.......)).]]]]]]].....  -26.3
```

The minimum-free-energy search uses the same pieces as the ensemble, but takes the best one instead of the sum. With a finite beam it keeps its own candidates.